Summary of Doc

The system is used by **four roles** in the organization.

1️⃣ Priya — Payment Operations Analyst

Frontline employee handling daily payments.

Responsibilities:

- Reviews AI suggestions
- Applies, holds, or escalates payments
- Handles **60–100 cases per day**

Example workflow:

1. Opens payment queue
2. Sees AI recommendation
3. Accepts or overrides it

Example:
AI says:

Apply payment
Confidence: 98%
Reason: exact name match + exact amount

Priya clicks **Apply**, and the payment is processed.

2️⃣ Damien — Investigation Specialist

Handles **hard cases that Priya cannot solve**.

Examples:

- No customer match
- Fraud risk
- Duplicate payments

His tasks:

- Investigate payment history
- Log if the sender was contacted
- Decide final outcome:
    - Apply payment
    - Return payment

    - Flag for fraud

3️⃣ Lorraine — Compliance / Operations Director

She **does not handle individual payments**.

Her role:

- Monitor system performance
- Ensure the AI decisions are safe and compliant
- Review metrics like:
    - How many payments were auto-applied
    - How often analysts override the AI
    - Escalation rates

She also **approves configuration changes** to the AI.

4️⃣ Marcus — System Admin / Data Analyst

Responsible for **maintaining and improving the AI system**.

His tasks:

- Monitor system analytics
- Adjust decision thresholds
- Investigate anomalies

Example:

If analysts are overriding the AI too often, he might change the threshold:

Old rule:
Auto apply if confidence ≥ 90%

New rule:
Auto apply if confidence ≥ 93%

But **Lorraine must approve the change first.**

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4. Typical Payment Processing Flow

Step 1: Payment arrives

Example:

Sender: John A. Smith
Amount: $2500
Policy number: missing

Step 2: AI analyzes the payment

Signals checked:

- Name similarity
- Amount variance
- Fraud flags
- Duplicate detection

Step 3: AI suggests an action

Example output:

Recommendation: HOLD
Confidence: 82%
Reason: Name match not strong enough

Step 4: Analyst decides

Priya can:

Accept recommendation
Override recommendation
Escalate

Step 5: Investigation (if escalated)

Damien investigates:

- Search system records

- Check payment patterns
- Log any Contact of sender

Step 6: Everything is logged

Every action is recorded in an **audit log**, including:

- AI decision
- Human decision
- Overrides
- Configuration changes
  user-flow-document

This is important for **regulatory compliance**.

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6. Why there are 4 roles

Each role serves a different purpose.

Role	Purpose
Analyst (Priya)	Process daily payments
Investigator (Damien)	Handle complex cases
Compliance Director (Lorraine)	Ensure governance and audits
System Admin (Marcus)	Improve AI performance

The document suggests that **in small teams, Damien and Marcus roles could be merged**, but in large companies all four roles are needed.

Overview

Overview

Problem being solved: Unidentified or miscellaneous payments enter the insurance payment system without sufficient reference data (missing policy numbers, mismatched amounts, unrecognised senders), landing in a suspense bucket that requires time-consuming manual investigation.

Solution approach: An AI agent analyses available signals (name similarity, amount variance, payment history, risk flags, duplicate detection) to infer the most likely customer match and recommend an action — Apply, Hold, or Escalate — with an explanation, confidence score, and complete audit trail.

Stakeholders mapped: Payment Operations Analyst (Priya), Payment Investigations Specialist (Damien), Operations Director / Compliance Lead (Lorraine), Business Systems Analyst / Ops Tech Lead (Marcus)

Total flows documented: 4 primary flows + cross-stakeholder dependency map

Key Dependencies Between Stakeholders

- Priya's escalation decisions feed Damien's queue
- Priya's override behaviour feeds Marcus's performance data
- Marcus's threshold configuration shapes every decision Priya and Damien see
- Lorraine's threshold approvals gate Marcus's configuration changes
- All four flows converge on the same underlying audit log

Stakeholder Index

Role / Goal	Connected To
Priya — Payment Operations Analyst Resolve daily payment queue accurately and efficiently using AI-assisted recommendations	Damien (escalation recipient), Marcus (override data source), Lorraine (SLA accountability)
Damien — Payment Investigations Specialist Investigate and close escalated high-risk cases using full upstream context	Priya (case sender), Lorraine (fraud/risk accountability), Marcus (pattern feedback source)
Lorraine — Operations Director / Compliance Lead Govern AI configuration, monitor system performance, and ensure audit-readiness	Marcus (configuration approver), Priya + Damien (downstream accountability)
Marcus — Business Systems Analyst / Ops Tech Lead	Lorraine (approval gatekeeper), Priya + Damien (feedback sources)

Configure, monitor, and continuously improve AI decision thresholds based on live performance data	

Shared Flows — All 4 Users

The following entry and authentication steps are identical for all four users. Divergence begins at the home screen, which renders differently per role.

Shared Step 1 — Authentication & Access

1. **Navigate** to the AI Payment Resolution Platform
2. **Authenticate** via SSO / corporate identity provider
3. **System validates** role and permissions
4. **System renders** the role-appropriate home screen — divergence begins here

Shared Step 2 — Audit Log (Background, All Roles)

Every action taken by any user — recommendations viewed, decisions made, overrides applied, thresholds changed, reports exported — is automatically written to the immutable system audit log in real time. No user initiates this; it runs as a system-layer behaviour behind all four flows.

Where Flows Differ vs. Remain the Same

What is Identical Across All Four Users

Shared Element	Detail
Authentication	Same SSO login, role assigned at auth time
Audit log	All actions automatically written to the same underlying audit log — all four users contribute to it
Case data model	All four users view the same underlying case data — role determines which fields and views are exposed
Notification system	All users receive role-appropriate in-app notifications from the same notification service
Export functionality	All roles can export data relevant to their scope (Priya: case detail; Lorraine: aggregate reports; Marcus: configuration history)

Where Flows Diverge by Role

Flow Point	Priya	Damien	Lorraine	Marcus
Home screen	Queue Dashboard (case list)	Investigation Queue (escalated only)	Governance Dashboard (aggregate metrics)	Admin Dashboard (per-scenario analytics)

Primary data unit	Individual payment case	Escalated case + handoff packet	Aggregate metric / config request	Performance trend / of AI model
Decision authority	Apply / Hold / Escalate within confidence bounds	Final determination on all escalated cases	Configuration approval	Configuration proposals (approved by Lorraine)
AI output consumed	Recommendation + confidence + signal values	Signal values + case handoff context	Aggregate AI decision distribution	Per-scenario performance metrics + override data
Audit log contribution	Case dispositions + overrides + annotations	Investigation determinations + contact records	Config approvals + review records	Config changes + anomaly investigation records
System config access	None	None	Approval-only (cannot edit directly)	Propose + deploy (post-approval)
Case access scope	Own queue cases	Escalated cases routed to investigation	Case summaries in exception report only	All

Cross-Stakeholder Dependency Map

From	Triggers	To	Condition	Impact if Broken
Priya	Escalation action	Damien's queue	Case cannot be resolved at analyst tier	Damien receives incomplete cases; investigation efficiency collapses
Priya	Override event	Marcus's analytics	Any override action	Marcus loses calibration signal; thresholds cannot be evidence-based
Damien	SLA expiry	Lorraine's exception dashboard	Case exceeds investigation SLA	Lorraine cannot make policy-level decisions; cases sit indefinitely
Damien	Resolution pattern data	Marcus's pattern analytics	Case closed with investigation record	Marcus cannot build cross-case pattern library
Marcus	Configuration change proposal	Lorraine for approval	Any threshold change	Without approval gate, governance integrity is lost
Lorraine	Configuration approval	Marcus for deployment	After approval	Without Lorraine's sign-off, Marcus cannot change production config
Lorraine	Metric anomaly flag	Marcus for investigation	Metric outside acceptable range	Threshold drift goes unaddressed; decision quality degrades
Marcus	Threshold configuration	All users (indirect)	All AI decisions use current thresholds	Miscalibrated thresholds degrade Priya's recommendations, Damien's context, and Lorraine's audit data simultaneously
System (audit log)	All action records	Lorraine's governance reports	Continuous background	If audit log fails, the entire compliance layer collapses across all four flows

Role Necessity Evaluation: Are 4 User Profiles Justified?

Assessment

Verdict: 3 profiles are essential. The 4th (Marcus) can be merged with conditions.

Priya (Analyst) — ESSENTIAL, cannot merge

Priya is the only role with transactional, case-level decision authority. Her workflow, data scope, and interaction patterns are fundamentally different from every other role. She is the primary user the product is designed for. Merging her with any other role would require a single interface to serve radically different mental models and task cadences. Non-negotiable separation.

Damien (Investigator) — ESSENTIAL at scale, conditionally mergeable

Damien's needs diverge from Priya's in three critical ways: (1) he requires a case handoff packet view that Priya doesn't produce; (2) his queue is scoped exclusively to escalated cases; (3) his decision authority is different — he makes final determinations, not routing decisions. However, in smaller organisations, Priya and Damien could be combined into a single "Senior Analyst" role where the same user handles both standard queue work and investigations. Viable consolidation only in small teams (1–3 analysts).

Lorraine (Compliance Director) — ESSENTIAL, cannot merge

Lorraine's profile exists for a fundamentally different reason than the other three: governance, not operations. Her data access is aggregate-only; she never touches individual cases. Her approval authority over configuration is a compliance control that must be enforced at the role level. Merging her with Marcus would eliminate the separation of duties between "proposes configuration changes" and "approves configuration changes" — a basic internal control principle in regulated industries. This separation is not optional in a financial services context.

Marcus (Sys Admin) — CONDITIONALLY MERGEABLE

Marcus is the most specialised of the four and the least frequently active. His tasks — threshold configuration, performance monitoring, anomaly diagnosis — could in a smaller deployment be handled by a technically proficient senior analyst (Priya-tier) with admin access added to their role, or by Lorraine herself if she has sufficient technical fluency and the team is small.

The case for keeping Marcus separate: as the organisation scales, the volume of configuration proposals, performance monitoring cycles, and anomaly investigations grows. Merging his responsibilities into another role creates cognitive overload and blurs the line between operational decision-making and system governance. In the PoC phase, a merged "Analyst + Admin" role might be acceptable. In production at scale, separation is warranted.

Final Recommendation

Profile	Keep Separate?	Merge Condition
Priya — Analyst	Always separate	No valid merge scenario
Damien — Investigator	Separate at scale	Merge with Priya-tier into "Senior Analyst" only in small teams (<3 analysts)

Lorraine — Compliance Director	Always separate	No valid merge — separation of duties is a compliance requirement
Marcus — Sys Admin	Conditionally	Merge with senior analyst acceptable in PoC / small-team deployment; separate at scale

Minimum viable profile set (PoC / small team): 3 profiles — Analyst (covers Priya + Damien tasks), Compliance Director (Lorraine), Sys Admin (Marcus, or merged into senior analyst)

Recommended production profile set (scale): 4 profiles as defined — each role has distinct enough access scope, decision authority, and interface needs to justify separation

Assumptions & Open Questions

Assumptions Made

- The 90% auto-apply threshold means Priya does not need to click "approve" for high-confidence cases — the system applies automatically and she sees the result
- "Supervisor approval" for Hold cases refers to a Priya-tier peer or team lead, not Lorraine (Lorraine operates at configuration level, not case level)
- Damien's investigation queue is automatically populated from escalations — he does not need to pull cases manually
- Marcus's override analytics are structured events captured by the system automatically — Priya does not need to "send" her overrides

Open Questions for Product / Research Validation

- Does "Hold requires approval" mean a second analyst reviews, or does Priya herself hold and await a system timeout before applying?
- Is there a defined SLA for Damien's investigation queue, and who sets it — Lorraine or an operational policy document?
- In the PoC deployment, is Marcus a dedicated role or is it a technical member of the ops team wearing two hats?
- What is the notification channel for policyholder outreach — in-system template, email, or phone only? This affects Dorothy's experience design.
- Does the staging / simulation environment need to be isolated from production data, or can it use anonymised production case history?

Payment Operations Analyst

FLOW 1 — Priya Venkataraman · Payment Operations Analyst

Context

Field	Detail
Who	Primary daily user; processes 60–100 unidentified payment cases per day
Primary goal	Resolve every case in queue to a disposition (Apply / Hold / Escalate) before close of business, using AI recommendations she can verify and trust
Entry point	Login → Queue Dashboard
Exit point / success state	All cases reach a disposition with a system-generated audit record; zero cases left in suspense without a status

Happy Path — Auto-Apply Case (Scenario 1, High Confidence)

1. **Login → Queue Dashboard loads**
   - System displays all open cases sorted by AI confidence score (highest first by default)
   - Each row shows: Scenario type, sender name, amount, AI recommendation, confidence band, age of case
2. **Priya selects a high-confidence case** flagged as Scenario 1, confidence 98%, recommendation: APPLY
3. **Case Detail View loads with full AI reasoning panel**
   - System displays: sender name, payment amount, policy reference, matched policyholder, name similarity score (100%), amount variance (0%), risk flags (none), policy status (Active)
   - AI recommendation panel: APPLY | Confidence 98% | "Perfect name match (100%), exact amount match, policy explicitly provided, no risk flags"
   - Decision point: Priya reviews and decides whether to accept or override
4. **Priya accepts the APPLY recommendation** — clicks "Accept & Apply"
   - Confidence ≥ 90%, no approval required per Scenario 1 rules
   - System: payment applied to policy; case status → "Resolved – Applied"; audit log entry written with all signal values, recommendation, action + timestamp
5. **Case closes; Priya returns to queue** — success state: case resolved, audit trail complete, queue decremented

Happy Path — Hold Case (Scenario 1, Medium Confidence / Risk Flag)

Steps 1–2 identical to above. Priya selects a case with confidence 82%, recommendation: HOLD.

6. **Case Detail View shows HOLD recommendation**
   - AI panel: HOLD | Confidence 82% | "Name match 82% — not strong enough for auto-apply; manual approval required"
   - OR: HOLD | Confidence 70% | "Perfect match but fraud history flag detected — manual review required"
7. **Priya reviews all signal values and agrees with HOLD** — clicks "Confirm Hold"

  - System: case → "Held – Pending Approval"; supervisor approval request generated; audit log updated
8. **Priya annotates the hold (optional but encouraged)**
  - Adds a brief case note, e.g. "Name variant — likely same person, but fraud flag requires review"
  - System: annotation saved to case record and audit log
9. **Case held; Priya returns to queue**

Happy Path — Escalate Case (Scenario 4 / Scenario 5)

Steps 1–2 identical. Priya selects a case with confidence 0%, recommendation: ESCALATE.

10. **Case Detail View shows ESCALATE recommendation**
  - Scenario 4: ESCALATE | Confidence 0% | "No matching customer was found. Best fuzzy match: [Name] at 72% — INACTIVE since [year]. Unable to determine customer intent."
  - Scenario 5: ESCALATE | Duplicate Probability 100% | "DUPLICATE ALERT: Matching payment PMT-504 detected within 72 hours. Policy $0 balance. Contact sender to confirm."
11. **Priya reviews and confirms escalation** — clicks "Escalate to Investigation"
  - System: case routed to Damien's investigation queue WITH full case context packet (all signals, AI reasoning, Priya's notes, action history)
  - Audit log updated; Priya's queue decremented
12. **Priya optionally adds pre-escalation notes** — these travel with the case to Damien

Alternative Paths

Alt Path 1 — Priya overrides an AI recommendation

- Trigger: Priya disagrees with the AI recommendation (e.g. AI says APPLY but she sees a signal it missed)
- Priya clicks "Override" → system presents override form: alternative action + mandatory reason field
- System applies override; records original AI recommendation, override action, and reason in audit log
- Override event also captured as a structured performance signal for Marcus's monitoring
- Outcome: case resolved per override; override data flows to admin performance dashboard

Alt Path 2 — Scenario 2 (Missing policy reference, disambiguation needed)

- Trigger: AI identifies customer but cannot determine which policy the payment belongs to
- Case Detail shows: HOLD | "Customer matched (100%). 2 active policies with identical premium. Cannot determine which policy — manual selection required."
- Outcome: payment applied to selected policy with approval flag, or held for further clarification

Alt Path 3 — Scenario 3 (Amount variance)

- Trigger: amount variance 2–15% (Hold) or >15% (Escalate)
- AI shows: "Payment $2,700 is 8% higher than expected $2,500 — Hold for review"
- Priya checks historical payment pattern (displayed in case view)

- If pattern consistent: annotates and accepts Hold. If >15%: confirms escalation to Damien

Alt Path 4 — Duplicate detected before Priya acts (Scenario 5)

- Trigger: Scenario 5 duplicate check fires before any other scenario evaluation
- Case enters queue already flagged: "POTENTIAL DUPLICATE — 100% match to PMT-504 within 72 hours"
- Priya checks outstanding balance: $0 = escalate; balance exists = Hold

Error States & Edge Cases

Scenario	Trigger	System Response	Recovery Path
AI recommendation missing	Signal computation failure	Case flagged "AI Unavailable — Manual Review Required"	Priya works case manually; all manual actions still logged
Policy inactive	Scenario 1, inactive policy status	System escalates to Investigator	
Override without reason	Priya submits override with no reason entered	Form validation error; submit blocked	Priya enters reason; override proceeds

Handoffs to Other Stakeholders

- Handoff to Damien: at escalation step — Priya's confirmation routes the case + full context packet to Damien's investigation queue
- Data to Marcus: every override Priya performs is captured as a structured performance event in the admin analytics layer
- Accountability to Lorraine: Priya's queue resolution rate and SLA adherence feed Lorraine's operational dashboard (automated, no Priya action required)

Payment Investigations Specialist

FLOW 2 — Damien Okafor · Payment Investigations Specialist

Context

Field	Detail
Who	Senior investigator who receives cases Priya cannot resolve — no-match escalations, fraud flags, confirmed duplicates
Primary goal	Close escalated cases with full investigative rigour, producing a defensible decision record and, where needed, coordinating with external parties
Entry point	Login → Investigation Queue (pre-filtered to escalated cases only)
Exit point / success state	Case closed with final determination (payment applied, returned, or policy actioned), full investigation record on file, policyholder contacted if needed

Happy Path — Scenario 4 (No Matching Customer)

1. **Login → Investigation Queue loads**
   - Cases displayed with: risk-level indicator, scenario type, escalation source (from Priya), time since escalation, amount, sender name
   - Queue pre-sorted by risk level — fraud-flagged cases surface first
2. **Damien opens case → Case Handoff Packet renders**
   - Upstream signals: name similarity score, max database similarity, amount correlation, policy reference search result
   - AI reasoning: full explanation text from the agent
   - Analyst history: Priya's actions, timestamps, any pre-escalation notes
3. **Damien builds a hypothesis and begins investigation**
   - Reviews all signals; cross-references amount against active policies; checks pending applications; reviews payment method
   - Decision point: can he identify the sender without external contact?
4. **If identifiable internally** — annotates case with determination; selects resolution action (Apply / Return / Flag policy)
   - System: case closed; determination and reasoning logged; audit record complete
5. **If external contact required** — system flags "Outreach Recommended"; Damien prepares from handoff packet
   - Damien contacts sender (phone or written) with full case context already in hand
   - Logs contact attempt + outcome in case annotation field
6. **Contact successful — Damien resolves case**
   - Sender provides clarification; Damien selects resolution action
   - System: case closed with contact record + determination logged
7. **Case closed; Damien returns to investigation queue** — success state: case resolved with full investigation record; audit trail complete

Happy Path — Scenario 5 (Duplicate Payment Investigation)

Steps 1–2 identical. Damien opens a Scenario 5 duplicate case.

8. **Handoff packet shows duplicate alert with full match detail**
   - System displays: original payment ID, amount, date/time, time delta (52 hours), 100% field match confirmation, outstanding balance ($0), next due date
   - AI reasoning: "DUPLICATE ALERT — contact sender to confirm intent or error"
9. **Damien contacts policyholder** — coordinates through single-outreach protocol (one contact owner assigned)
   - Decision point: confirmed duplicate vs. confirmed intentional second payment
10. **Confirmed duplicate** — flags case for payment reversal; coordination with payment method provider begins; annotates with contact record
11. **Confirmed intentional** (e.g. catch-up payment) — overrides duplicate flag with documented justification; applies payment to policy
12. **Case closed with full investigation + contact record**

Alternative Paths

Alt Path 1 — Pattern recognised from prior investigation

- Trigger: Damien recognises the signal combination from a previous case (e.g. business-name sender + invoice reference + personal policy)
- Searches cross-case pattern library by signal type or sender pattern; finds precedent case
- Applies same resolution logic with documentation referencing the precedent
- Outcome: faster resolution; precedent linkage captured in case record

Alt Path 2 — No response from sender, case cannot be closed

- Trigger: Damien initiates outreach; no response after defined follow-up attempts
- Damien logs all contact attempts with timestamps
- Case moves to "Pending Sender Response" status with a defined SLA timer
- On SLA expiry: case escalated to Lorraine's exception report for policy-level decision
- Outcome: case escalated to governance tier; Lorraine sees it in her exception dashboard

Alt Path 3 — New policy application found (Scenario 4, edge case)

- Trigger: Damien searches system and finds a pending policy application matching sender details
- Annotates case: "Pending application found — [reference]"
- Coordinates with underwriting team (outside this system) to confirm
- Routes payment to appropriate holding account pending policy activation

Error States & Edge Cases

Scenario	Trigger	System Response	Recovery Path
Case arrives without analyst context	Priya's notes missing from handoff packet	Warning: "Analyst context incomplete — some upstream data may be unavailable"	Damien proceeds manually; flags gap to Marcus via anomaly report

Contact attempt cannot be logged	System annotation save fails	Inline save error with retry prompt	Damien retries; if persistent, logs manually and flags IT
Duplicate reversal coordination fails	Payment method provider rejects reversal request	System flags "Reversal Failed — Manual Coordination Required"	Damien escalates to finance team; case remains open with flag

Handoffs to Other Stakeholders

- Receives from Priya: at case entry — full case handoff packet including all AI signals, analyst action history, and pre-escalation notes
- Data to Marcus: Damien's case annotations and resolution patterns feed the cross-case pattern library and investigation outcome data
- Escalation to Lorraine: cases that exceed SLA or require policy-level decisions appear in Lorraine's exception report (automated trigger)
- Outbound to Policyholder: single coordinated contact per case; contact record logged and attached to case

Compliance Lead

FLOW 3 — Lorraine Thibodeau · Operations Director / Compliance Lead

Context

Field	Detail
Who	Executive sponsor; never works individual cases; responsible for configuration governance, compliance documentation, and operational oversight
Primary goal	Ensure the AI system's decisions are auditable, the configuration is defensible, and performance metrics confirm the system is operating within acceptable risk parameters
Entry point	Login → Governance Dashboard (aggregate views only — no case-level access)
Exit point / success state	Configuration approved and documented; audit reports export-ready; performance metrics confirm system health; no open compliance exceptions

Happy Path — Weekly Performance Review

1. **Login → Governance Dashboard loads**
   - Displays: decision distribution summary (Apply/Hold/Escalate counts and %), override rate trend, SLA adherence rate, escalation volume trend, confidence score histogram
   - Date range selector defaults to "this week"
   - Decision point: are any metrics outside acceptable range?
2. **Metrics within range — Lorraine reviews and records review**
   - Clicks "Record Review" — system timestamps her review of the period's report
   - System: review recorded in governance log
3. **Review complete** — success state: period reviewed, report exported, governance log updated

Happy Path — Configuration Change Approval

4. **Lorraine receives a configuration change request from Marcus**
   - Entry: email notification OR in-platform notification — "Configuration Change Request Pending Approval"
5. **Lorraine opens the change request**
   - System displays: parameter name, current value, proposed value, Marcus's rationale, supporting performance data, projected impact (from back-test)
   - Decision point: approve, reject, or request more information
6. **Lorraine approves** — clicks "Approve Change" + optional comment
   - System: configuration updated in production; change recorded in version-controlled log with Lorraine's identity, timestamp, and comment
   - Marcus notified of approval
7. **Lorraine rejects or requests more data** — clicks "Reject" or "Request More Information" + mandatory comment

  - System: change request returned to Marcus with comment; configuration unchanged

Alternative Paths

Alt Path 1 — Anomaly detected in metrics

- Trigger: escalation rate spikes outside normal band; override rate anomalously high; auto-apply rate drops significantly
- Dashboard metric cell highlighted in amber or red
- Lorraine drills into the anomalous metric to identify which scenario type is driving it
- Opens a "Flag for Investigation" request to Marcus — pre-populated with the anomalous metric, scenario type, and date range
- Outcome: investigation triggered; Lorraine's flag recorded in governance log

Alt Path 2 — Audit request from external regulator

- Trigger: regulatory body requests documentation of AI decision logic and decision records for a specific period
- Lorraine navigates to Audit Export view; selects date range and export scope
- System generates structured export: decision records, configuration version history with rationale, override analysis
- Lorraine reviews export for completeness before submitting
- Outcome: audit-ready export produced; submission documented in governance log

Alt Path 3 — Damien SLA exception surfaces

- Trigger: a case in Damien's queue exceeds the defined investigation SLA with no sender response
- Exception appears in Lorraine's dashboard with case ID and days without resolution
- Lorraine reviews case summary (not full case detail — summary only)
- Decision: authorise payment return, authorise indefinite hold, or escalate to legal/compliance team
- Lorraine records decision with rationale; Damien receives instruction to action

Error States & Edge Cases

Scenario	Trigger	System Response	Recovery Path
Audit export fails	System error during report generation	Error notification with retry option	Lorraine retries; if persistent, IT support escalation
Undocumented config change	System detects configuration parameter changed without formal request	Alert: "Untracked configuration change detected"	Immediate investigation flag to Marcus; change frozen pending review
Dashboard data delayed	Reporting pipeline lag	Banner: "Data last updated [timestamp] — refresh in progress"	Lorraine waits for refresh; critical decisions deferred until data is current

Handoffs to Other Stakeholders

- Approves from Marcus: configuration change requests — Lorraine's approval is required before any threshold change takes effect
- Monitors Priya + Damien: queue SLA, override rates, and escalation volumes feed her dashboard automatically — no direct interaction required
- Receives exception from Damien: SLA-expired cases surface in her exception dashboard for policy-level decision
- Triggers Marcus: when she flags a metric anomaly, the flag creates an investigation task for Marcus

Analyst / Ops Tech Lead

FLOW 4 — Marcus Sundaram · Business Systems Analyst / Ops Tech Lead

Context

Field	Detail
Who	System owner for configuration and performance; not a daily transaction user; accesses the system periodically for monitoring and reactively when anomalies are reported
Primary goal	Keep the AI decision thresholds calibrated to real operational patterns; maintain performance data; diagnose anomalies; propose and implement approved configuration improvements
Entry point	Login → Admin Dashboard OR reactive entry from an anomaly alert / Lorraine's flag
Exit point / success state	Thresholds are calibrated; performance data is clean; configuration changes are approved and deployed; anomalies are explained and resolved

Happy Path — Periodic Performance Review & Threshold Calibration

1. **Login → Admin Dashboard loads**
   - Displays per-scenario analytics: case volume by scenario type (trend over time), decision distribution per scenario (Apply/Hold/Escalate %), override rate per scenario and confidence band, confidence score histogram, override reason analysis
   - Decision point: are any metrics signalling threshold drift?
2. **Marcus identifies a leading indicator of miscalibration**
   - Example: override rate in Scenario 1 at 90–92% confidence band has increased from 3% to 11% over 4 weeks
   - Signals: the 90% auto-apply threshold may be too permissive — the AI is approving cases Priya is overriding
3. **Marcus reviews override reason data**
   - Override Analysis View: filterable by scenario, confidence band, date range, override reason category
   - Finds: most overrides at 90–92% band annotated "name format variation — not confident enough"
   - Decision point: is a threshold change warranted?
4. **Marcus builds a change proposal**
   - Opens Configuration Change Request form: parameter name, current value, proposed value, rationale, supporting evidence, projected impact
   - Runs back-test: simulates raising auto-apply threshold to 93% against last 30 days of cases
   - Submits proposal to Lorraine for approval
5. **Lorraine approves** — see Lorraine's Flow, Happy Path 2, Step 3a
6. **Marcus deploys the approved change**
   - Opens Configuration Management view; clicks "Deploy to Production"
   - System: threshold updated; new version created in configuration history with parameter, old/new values, rationale, approval, deployment timestamp

7. **Marcus monitors post-change performance**
   - Watches override rate for the affected band over the following 1–2 weeks
   - Success state: override rate returns to baseline; configuration change documented and in effect

Happy Path — Anomaly Diagnosis (Reactive Entry)

8. **Marcus receives an anomaly alert**
   - Entry: system alert OR Lorraine's "Flag for Investigation" task OR Priya/Damien reporting unexpected behaviour
   - Alert contains: scenario type, affected metric, date range, case IDs exhibiting anomalous routing
9. **Marcus opens the Decision Trace View for an affected case**
   - Screen: Case Decision Trace (admin-only view)
   - Plain-language trace: "Payment received → Scenario 5 duplicate check: no match found → Scenario 1 evaluation: name similarity 91% (above 90% threshold) → amount variance 1.8% (within 2% tolerance) → risk flags: none → RECOMMENDATION: APPLY"
   - Decision point: does the trace reveal a logic error, data error, or threshold miscalibration?
10. **Logic / threshold issue identified** — Marcus notes which threshold produced the unexpected result; opens Configuration Change Request (see Happy Path above, steps 4–7)
11. **Data quality issue identified** — Marcus raises a data quality flag to the relevant upstream system owner; documents finding and impact in the anomaly record
12. **Anomaly resolved and documented** — root cause and resolution recorded; success state: anomaly explained, no recurring unexplained behaviour

Alternative Paths

Alt Path 1 — Emergency threshold rollback

- Trigger: a deployed configuration change produces clearly negative outcomes (e.g. escalation rate doubles overnight)
- Marcus detects acute metric deterioration on Admin Dashboard
- Opens Configuration Management view → clicks "Roll Back to Previous Version" — requires Lorraine's emergency approval (expedited in-system flow)
- Previous configuration restored; incident documented
- Outcome: configuration restored to previous state; incident record created for post-mortem

Alt Path 2 — Lorraine flags a metric anomaly (reactive)

- Trigger: Lorraine's "Flag for Investigation" task arrives in Marcus's notification queue
- Marcus opens the flag — pre-populated with the anomalous metric, scenario type, and date range
- Proceeds to anomaly diagnosis flow (Happy Path above, from Step 2)

Alt Path 3 — Staging / simulation for a proposed change

- Trigger: Marcus wants to validate a proposed threshold change before submitting to Lorraine
- Opens Staging / Simulation view; inputs proposed parameter value; selects historical case set (e.g. last 30 days)

- System runs simulation: shows projected decision distribution change (how many cases would shift from Escalate → Hold → Apply)
- Marcus reviews projected impact, adjusts proposal if needed, attaches simulation output to change request

Error States & Edge Cases

Scenario	Trigger	System Response	Recovery Path
Decision trace unavailable	Logging gap for a specific case	Warning: "Decision trace incomplete — partial data only"	Marcus investigates logging pipeline; flags as data quality issue
Configuration deployment fails	System error during production update	Rollback initiated automatically; alert sent to Marcus	Marcus investigates and retries; Lorraine notified if delay exceeds SLA
Simulation produces no data	Insufficient historical cases in selected range	Warning: "Insufficient data — select a wider date range"	Marcus expands date range or notes limitation in change proposal

Handoffs to Other Stakeholders

- Submits to Lorraine: every configuration change proposal — requires her approval before deployment
- Receives from Lorraine: metric anomaly investigation flags; emergency rollback approvals
- Receives from Priya (indirect): override events are captured as structured data feeding Marcus's analytics — automatic system capture, no Priya action required
- Receives from Damien (indirect): investigation outcomes and case resolution patterns feed cross-case analytics — same automated capture mechanism